Physics
Standard and higher level
Paper 1A and Paper 2
B.3 Gas laws

12 October 2026

W4ss Academy practice test 1 | **Topic** B.3 Gas laws | **Ref.** W4SS–PHY–B3–01

Student name

30 minutes [Paper 1A and Paper 2]

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- **Paper 1A:** for each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.
- **Paper 2:** answers must be written within the answer boxes provided.
- A calculator is required for this paper.
- A clean copy of the **physics data booklet** is required for this paper.
- The maximum mark for paper 1A is **[10 marks]**.
- The maximum mark for paper 2 is **[10 marks]**.
- The maximum mark for this examination paper is **[20 marks]**.

Paper 1A

1. A substance in the gas state has a density about 1000 times less than when it is in the liquid state. The diameter of a molecule is d. What is the best estimate of the average distance between molecules in the gas state?

A. d

B. $10d$

C. $100d$

D. $1000d$

2. Under which conditions of pressure and density will a real gas approximate to an ideal gas?

	Pressure	Density
A.	high	high
B.	high	low
C.	low	high
D.	low	low

3. An ideal gas of constant mass is heated in a container of constant volume.

What is the reason for the increase in pressure of the gas?

A. The average number of molecules per unit volume increases.

B. The average force per impact at the container wall increases.

C. Molecules collide with each other more frequently.

D. Molecules occupy a greater fractional volume of the container.

4. A fixed mass of an ideal gas has a volume of V, a pressure of p and a temperature of 30 °C. The gas is compressed to the volume of $\frac{V}{6}$ and its pressure increases to $12p$. What is the new temperature of the gas?

A. 15 °C

B. 60 °C

C. 333 °C

D. 606 °C

5. Three statements about Boltzmann's constant k_B are:

I. k_B has a unit of $J\,K^{-1}$

II. $k_B = \dfrac{\text{gas constant}}{\text{Avogadro's constant}}$

III. $k_B = \dfrac{\text{the average kinetic energy of particles}}{\text{temperature of the gas}}$

Which statements are correct?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

6. Two ideal gases X and Y are at the same temperature. The mass of a particle of gas X is larger than the mass of a particle of gas Y. Which is correct about the average kinetic energy and the average speed of the particles in gases X and Y?

	Average kinetic energy	**Average speed**
A.	larger for Y	larger for Y
B.	same	larger for Y
C.	same	same
D.	larger for Y	same

7. A gas storage tank of fixed volume V contains N molecules of an ideal gas at temperature T. The pressure at kelvin temperature T is 20 MPa. $\frac{N}{4}$ molecules are removed and the temperature changed to $2T$. What is the new pressure of the gas?

A. 10 MPa

B. 15 MPa

C. 30 MPa

D. 40 MPa

8. A quantity of 2.00 mol of an ideal gas is maintained at a temperature of 127 ºC in a container of volume 0.083 m^3. What is the pressure of the gas?

A. 8 kPa

B. 25 kPa

C. 40 kPa

D. 80 kPa

9. A container holds 20 g of argon-40($^{40}_{18}Ar$) and 40 g of neon-20 ($^{20}_{10}Ne$) .

What is $\frac{\text{number of atoms of argon -40}}{\text{number of atoms of neon -20}}$ in the container?

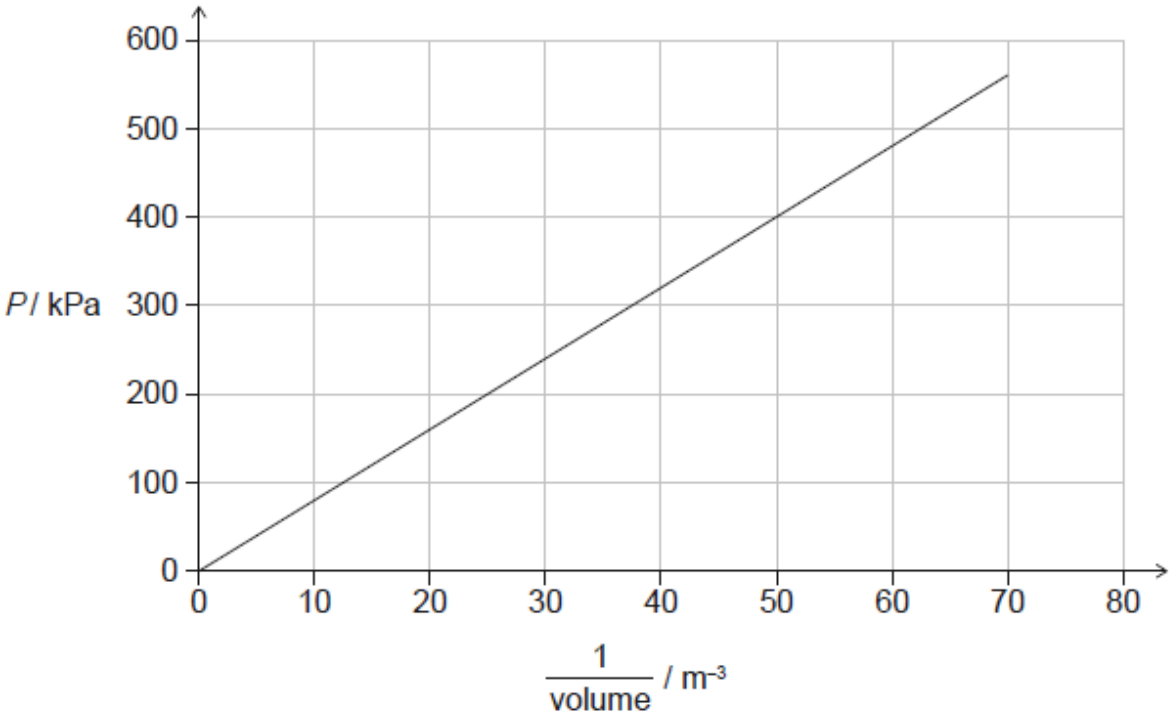
A. 0.25

B. 0.5

C. 2

D. 4

10. An ideal gas is maintained at a temperature of 100 K. The variation of the pressure P and $\frac{1}{\text{volume}}$ of the gas is shown.



What is the quantity of the gas?

A. $\frac{2 \times 10^5}{R}$ mol

B. $\frac{200}{R}$ mol

C. $\frac{80}{R}$ mol

D. $\frac{4}{5R}$ mol

Paper 1A — answer sheet

Student name _____ Date _____

1.	A	B	C	D
2.	A	B	C	D
3.	A	B	C	D
4.	A	B	C	D
5.	A	B	C	D
6.	A	B	C	D
7.	A	B	C	D
8.	A	B	C	D
9.	A	B	C	D
10.	A	B	C	D

Shade **one** circle per question. If you change your answer, cross out the wrong circle clearly.

Paper 2

Answer **all** questions. Answers must be written within the answer boxes provided.

1. A fixed mass of an ideal gas is contained in a cylinder closed with a frictionless piston. The volume of the gas is 2.5×10^{-3} m^3 when the temperature of the gas is 37 °C and the pressure of the gas is 4.0×10^5 Pa.

Energy is now supplied to the gas and the piston moves to allow the gas to expand. The temperature is held constant.

(a) Calculate the number of gas particles in the cylinder. [2]

(b) (i) Discuss, for this process, the changes that occur in the density of the gas. [2]

(ii) Discuss, for this process, the changes that occur in the internal energy of the gas. [2]

2. A container holds a mixture of argon and helium atoms at a temperature of 37 °C.

Calculate the average translational speed of the argon atoms.

The molar mass of argon is 4.0×10^{-2} kg mol $^{-1}$. [4]

Sources of the questions

This practice test was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Physics Questionbank (first assessment 2025), topic B.3 Gas laws. They remain the property of the International Baccalaureate Organization and must not be published or sold.

Paper 1A — question 1	20N.1A.SL.TZ0.12
Paper 1A — question 2	19N.1A.SL.TZ0.9
Paper 1A — question 3	20N.1A.SL.TZ0.11
Paper 1A — question 4	21N.1A.SL.TZ0.12
Paper 1A — question 5	22N.1A.SL.TZ0.10
Paper 1A — question 6	21M.1A.SL.TZ1.10 (variant 2)
Paper 1A — question 7	19M.1A.SL.TZ1.13
Paper 1A — question 8	21M.1A.SL.TZ1.12
Paper 1A — question 9	19M.1A.SL.TZ2.12
Paper 1A — question 10	22M.1A.SL.TZ1.12
Paper 2 — question 1	22M.2.SL.TZ2.2
Paper 2 — question 2	EXE.2.SL.TZ0.9 (part (b))

Please **do not** write on this page.

Answers written on this page
will not be marked.